

THE PHYSICAL AI KIT: END-TO-END DUAL-BRAIN ARCHITECTURE

Heterogeneous Multi-Rate Pipeline · Untrusted Linux MPU Proposals $\longleftrightarrow$ Trusted Real-Time MCU Permission

HOST BRAIN: LINUX MPU (Cognitive Cortex)

0.5–60 Hz · Best-Effort Scheduling · Untrusted Proposals

Ingestion & Spatial Tokenizer (60 Hz)

MIPI CSI-2 DMA Rings · DINOv2 / MobileNet → 3D Spatial Affordance Tokens

Temporal World Model & Frame Graph (50 Hz)

SE(3) Lie Algebra Tree · Proprioceptive Innovation Gate · Latent JEPA Belief

Semantic Reasoning & Deliberation (1–2 Hz)

Edge VLM (e.g. PaliGemma / SmolVLM) → 3D Bounding Goal · Expiring Lease t_{expire}

Trajectory Planning & Action Chunking (20 Hz)

Diffusion Policy / ACT → $H = 16$ Step Action Chunks · C^2 Hermite Spline Ensembling

REFLEX BRAIN: REAL-TIME MCU (Safety Enforcer)

1000 Hz Hard Real-Time Loop · Deterministic · Zero Malloc

Control Barrier Invariants ($h(x) \geq 0$)

Zero-Allocation 1 kHz QP · Workspace Geofence · Payload & Force Bounds

Dynamic Stopping Physics (d_{stop})

$$d_{\text{stop}}(t) = v(t) \cdot t_{\text{delay}} + \frac{v(t)^2}{2a_{\text{max}}} \leq d_{\text{clearance}} \cdot \text{Veto Accelerations}$$

Heartbeat Supervisor & Watchdog Lease

Hardware Countdown Timer · $\tau_{\text{watchdog}} \leq 50 \text{ ms}$ · Trips on Linux Crash / Hang

ISO 13850 Fallback Escalation FSM

Cat 2 (Position Hold) → Cat 1 (Dynamic Brake) → Cat 0 (Power Relay Open)

~~THE PROPOSAL – PERMISSION PRIVILEGE BOUNDARY (MPU Proposes p_t $\longleftrightarrow$ MCU Permits u_t)~~

HETEROGENEOUS Tightly-Coupled Memory (TCM) Shared SRAM (Lock-Free Seqlocks & DMA Rings)

→ **Proposals (MPU → MCU):** Expiring Intent Leases (t_{expire}), H -step Action Splines (p_t), Model Confidence (κ)

← **Telemetry (MCU → MPU):** Encoders, Voltages, Veto Flags, PTP Hardware Timestamps

SENSOR SUITE (Environment Ingestion)

MIPI CSI-2 Vision · 6-DoF IMU · Shunt Sense · PTP Time Base

ACTUATION STAGE & SAFETY RELAYS

3-Phase Gate Drivers · BLDC Motor Stage · Emergency Cutoff Relay

THE PHYSICAL WORLD ($W_t \longrightarrow W_{t+1}$) · Kinetic Momentum

($p = mv$) · Joule Heat (I^2R) · Contact Dynamics · **Zero Software Undo**